```
// ==================== cloudfunctions/admin/index.js ====================
const cloud = require('wx-server-sdk')
cloud.init({ env: cloud.DYNAMIC_CURRENT_ENV })
const db = cloud.database()
const users = db.collection('users')

// 获取当前用户（网页 token 优先，其次小程序 openid）
async function getCurrentUser(event) {
  const { OPENID } = cloud.getWXContext()
  if (event.token) {
    const res = await users.where({ token: event.token }).get()
    return res.data.length > 0 ? res.data[0] : null
  }
  if (OPENID) {
    const res = await users.where({ _openid: OPENID }).get()
    return res.data.length > 0 ? res.data[0] : null
  }
  return null
}

exports.main = async (event, context) => {
  const action = event.action || 'listUsers'

  try {
    const user = await getCurrentUser(event)
    if (!user || user.isAdmin !== true) {
      return { code: 2, message: '无权限：仅管理员可操作' }
    }

    switch (action) {
      case 'listUsers': {
        const res = await users.orderBy('createTime', 'asc').get()
        const list = res.data.map(u => ({
          _id: u._id,
          name: u.name,
          isAdmin: u.isAdmin === true,
          avatarFileID: u.avatarFileID || '',
          createTime: u.createTime
        }))
        return { code: 0, list }
      }

      case 'setAdmin': {
        const { userId, isAdmin: target } = event
        if (!userId) return { code: 1, message: '缺少用户 ID' }

        // 不能取消自己的管理员权限
        if (user._id === userId && !target) {
          return { code: 1, message: '不能取消自己的管理员权限' }
        }

        await users.doc(userId).update({ data: { isAdmin: target === true } })
        return { code: 0, message: target ? '已设为管理员' : '已取消管理员' }
      }

      case 'deleteUser': {
        const { userId } = event
        if (!userId) return { code: 1, message: '缺少用户 ID' }

        // 不能删除自己
        if (user._id === userId) {
          return { code: 1, message: '不能移除自己' }
        }
```

```
        await users.doc(userId).remove()
        return { code: 0, message: '已移除该用户' }
      }

      default:
        return { code: 1, message: '未知操作' }
    }
  } catch (err) {
    console.error(err)
    return { code: 1, message: '服务异常', error: err.message }
  }
}

// ==================== cloudfunctions/borrow/index.js ====================
const cloud = require('wx-server-sdk')
cloud.init({ env: cloud.DYNAMIC_CURRENT_ENV })
const db = cloud.database()
const equipmentCol = db.collection('equipment')
const recordsCol = db.collection('borrow_records')
const users = db.collection('users')

// 获取用户（网页 token 优先，其次小程序 openid）
async function getUser(event) {
  const { OPENID } = cloud.getWXContext()
  if (event.token) {
    const res = await users.where({ token: event.token }).get()
    return res.data.length > 0 ? res.data[0] : null
  }
  if (OPENID) {
    const res = await users.where({ _openid: OPENID }).get()
    return res.data.length > 0 ? res.data[0] : null
  }
  return null
}

exports.main = async (event, context) => {
  const { OPENID } = cloud.getWXContext()
  const action = event.action || 'records'
  const token = event.token || ''
  const isWeb = !!token // 网页版一定有 token，小程序没有

  try {
    switch (action) {
      case 'borrow': {
        const user = await getUser(event)
        if (!user) return { code: 2, message: '请先登录' }

        const { id } = event
        if (!id) return { code: 1, message: '缺少器材 ID' }

        const eqRes = await equipmentCol.doc(id).get().catch(() => null)
        if (!eqRes || !eqRes.data) return { code: 1, message: '器材不存在' }
        const eq = eqRes.data
        if (eq.status === 'borrowed') {
          return { code: 1, message: `该器材已被 ${eq.currentBorrower || '他人'} 借走
` }
        }

        const now = db.serverDate()
        const reason = (event.reason || '').trim()
        await equipmentCol.doc(id).update({
          data: {
            status: 'borrowed',
            currentBorrower: user.name,
            currentBorrowerOpenid: isWeb ? '' : OPENID,
```

```
        currentBorrowerToken: isWeb ? token : '',
        borrowTime: now,
        currentReason: reason
      }
    })

    await recordsCol.add({
      data: {
        equipmentId: id,
        equipmentName: eq.name,
        equipmentCode: eq.code || '',
        borrowerName: user.name,
        borrowerOpenid: isWeb ? '' : OPENID,
        borrowerToken: isWeb ? token : '',
        borrowTime: now,
        reason,
        returnTime: null,
        returnImageFileID: '',
        status: 'borrowed'
      }
    })

    return { code: 0, message: '借用成功' }
  }

  case 'return': {
    const user = await getUser(event)
    if (!user) return { code: 2, message: '请先登录' }

    const { id } = event
    if (!id) return { code: 1, message: '缺少器材 ID' }

    const eqRes = await equipmentCol.doc(id).get().catch(() => null)
    if (!eqRes || !eqRes.data) return { code: 1, message: '器材不存在' }
    const eq = eqRes.data

    if (eq.status === 'available') {
      return { code: 1, message: '该器材已在器材室，无需归还' }
    }

    // 借用人本人（网页 token 或小程序 openid）或管理员
    const isBorrower = isWeb
      ? eq.currentBorrowerToken === token
      : eq.currentBorrowerOpenid === OPENID
    const isAdminUser = user.isAdmin === true
    if (!isBorrower && !isAdminUser) {
      return { code: 2, message: '只能由借用人本人或管理员归还' }
    }

    const now = db.serverDate()
    await equipmentCol.doc(id).update({
      data: {
        status: 'available',
        currentBorrower: '',
        currentBorrowerOpenid: '',
        currentBorrowerToken: '',
        borrowTime: null,
        currentReason: ''
      }
    })

    const recRes = await recordsCol.where({
      equipmentId: id,
      status: 'borrowed'
    }).orderBy('borrowTime', 'desc').limit(1).get()
```

```
        if (recRes.data.length > 0) {
          await recordsCol.doc(recRes.data[0]._id).update({
            data: {
              status: 'returned',
              returnTime: now,
              returnImageFileID: event.returnImageFileID || ''
            }
          })
        }

        return { code: 0, message: '归还成功' }
      }

      case 'records': {
        const user = await getUser(event)
        if (!user) return { code: 2, message: '请先登录' }

        let res
        if (user.isAdmin === true) {
          res = await recordsCol.orderBy('borrowTime', 'desc').limit(100).get()
        } else if (isWeb) {
          res = await recordsCol.where({ borrowerToken:
token }).orderBy('borrowTime', 'desc').limit(100).get()
        } else {
          res = await recordsCol.where({ borrowerOpenid:
OPENID }).orderBy('borrowTime', 'desc').limit(100).get()
        }
        return { code: 0, list: res.data }
      }

      default:
        return { code: 1, message: '未知操作' }
    }
  } catch (err) {
    console.error(err)
    return { code: 1, message: '服务异常', error: err.message }
  }
}

// ==================== cloudfunctions/equipment/index.js ====================
const cloud = require('wx-server-sdk')
cloud.init({ env: cloud.DYNAMIC_CURRENT_ENV })
const db = cloud.database()
const equipmentCol = db.collection('equipment')
const users = db.collection('users')

// 判断是否是管理员（token 优先，其次 openid）
async function isAdmin(event) {
  const { OPENID } = cloud.getWXContext()
  if (event.token) {
    const res = await users.where({ token: event.token }).get()
    return res.data.length > 0 && res.data[0].isAdmin === true
  }
  if (OPENID) {
    const res = await users.where({ _openid: OPENID }).get()
    return res.data.length > 0 && res.data[0].isAdmin === true
  }
  return false
}

exports.main = async (event, context) => {
  const action = event.action || 'list'

  try {
```

```
    switch (action) {
      case 'list': {
        // 所有用户（含未登录）都可以查看器材列表
        const res = await equipmentCol.orderBy('createTime', 'desc').get()
        return { code: 0, list: res.data }
      }

      case 'detail': {
        const { id } = event
        if (!id) return { code: 1, message: '缺少器材 ID' }
        const res = await equipmentCol.doc(id).get()
        return { code: 0, item: res.data }
      }

      case 'add': {
        if (!(await isAdmin(event))) return { code: 2, message: '无权限：仅管理员可添加
器材' }

        const name = (event.name || '').trim()
        if (!name) return { code: 1, message: '请填写器材名称' }

        if (event.code) {
          const dup = await equipmentCol.where({ code: event.code }).get()
          if (dup.data.length > 0) return { code: 1, message: '该编号已存在' }
        }

        const addRes = await equipmentCol.add({
          data: {
            name,
            category: event.category || '机身',
            code: (event.code || '').trim(),
            brand: (event.brand || '').trim(),
            imageFileID: event.imageFileID || '',
            remark: (event.remark || '').trim(),
            status: 'available',
            currentBorrower: '',
            currentBorrowerOpenid: '',
            currentBorrowerToken: '',
            currentReason: '',
            borrowTime: null,
            createTime: db.serverDate()
          }
        })
        return { code: 0, id: addRes._id, message: '添加成功' }
      }

      case 'update': {
        if (!(await isAdmin(event))) return { code: 2, message: '无权限：仅管理员可编辑
器材' }

        const { id } = event
        if (!id) return { code: 1, message: '缺少器材 ID' }

        const updateData = {}
        if (event.name !== undefined) updateData.name = (event.name || '').trim()
        if (event.category !== undefined) updateData.category = event.category
        if (event.code !== undefined) updateData.code = (event.code || '').trim()
        if (event.brand !== undefined) updateData.brand = (event.brand || '').trim()
        if (event.imageFileID !== undefined) updateData.imageFileID =
event.imageFileID
        if (event.remark !== undefined) updateData.remark = (event.remark ||
'').trim()

        if (updateData.code) {
```

```
          const dup = await equipmentCol.where({ code: updateData.code }).get()
          if (dup.data.length > 0 && dup.data[0]._id !== id) {
            return { code: 1, message: '该编号已存在' }
          }
        }

        await equipmentCol.doc(id).update({ data: updateData })
        return { code: 0, message: '修改成功' }
      }

      case 'delete': {
        if (!(await isAdmin(event))) return { code: 2, message: '无权限：仅管理员可删除
器材' }

        const { id } = event
        if (!id) return { code: 1, message: '缺少器材 ID' }
        await equipmentCol.doc(id).remove()
        return { code: 0, message: '删除成功' }
      }

      default:
        return { code: 1, message: '未知操作' }
    }
  } catch (err) {
    console.error(err)
    return { code: 1, message: '服务异常', error: err.message }
  }
}

// ==================== cloudfunctions/init/index.js ====================
const cloud = require('wx-server-sdk')
cloud.init({ env: cloud.DYNAMIC_CURRENT_ENV })
const db = cloud.database()

exports.main = async () => {
  const collections = ['users', 'equipment', 'borrow_records']
  const results = []

  for (const name of collections) {
    try {
      await db.createCollection(name)
      results.push({ name, status: 'created' })
    } catch (e) {
      results.push({ name, status: 'exists_or_failed', msg: e.message })
    }
  }

  return { code: 0, results }
}

// ==================== cloudfunctions/login/index.js ====================
const cloud = require('wx-server-sdk')
const crypto = require('crypto')
cloud.init({ env: cloud.DYNAMIC_CURRENT_ENV })
const db = cloud.database()
const users = db.collection('users')

const SALT = 'camera-equipment-manager-salt'

function hashPassword(name, password) {
  return crypto.createHash('sha256').update(name + ':' + password + ':' +
SALT).digest('hex')
}
```

```
function genToken() {
  return crypto.randomBytes(24).toString('hex')
}

function cleanUser(user) {
  return {
    _id: user._id,
    _openid: user._openid || '',
    name: user.name,
    isAdmin: user.isAdmin === true,
    avatarFileID: user.avatarFileID || ''
  }
}

exports.main = async (event, context) => {
  const { OPENID } = cloud.getWXContext()
  const action = event.action || 'login'

  try {
    switch (action) {
      // ---------- 小程序：静默登录 ----------
      case 'login': {
        if (!OPENID) return { code: 1, message: '无法获取微信身份' }
        const res = await users.where({ _openid: OPENID }).get()
        if (res.data.length === 0) return { code: 0, isNew: true }
        return { code: 0, isNew: false, userInfo: cleanUser(res.data[0]) }
      }

      // ---------- 小程序：注册 ----------
      case 'register': {
        if (!OPENID) return { code: 1, message: '无法获取微信身份' }
        const name = (event.name || '').trim()
        if (!name) return { code: 1, message: '请填写姓名' }

        const exist = await users.where({ _openid: OPENID }).get()
        if (exist.data.length > 0) {
          return { code: 1, message: '该微信已注册', userInfo:
cleanUser(exist.data[0]) }
        }

        const countRes = await users.count()
        const isAdmin = countRes.total === 0

        const addRes = await users.add({
          data: {
            _openid: OPENID,
            name,
            isAdmin,
            avatarFileID: '',
            passwordHash: '',
            token: '',
            createTime: db.serverDate()
          }
        })
        return {
          code: 0,
          isNew: false,
          userInfo: { _id: addRes._id, _openid: OPENID, name, isAdmin, avatarFileID:
'' }
        }
      }

      // ---------- 网页：注册 ----------
      case 'webRegister': {
        const name = (event.name || '').trim()
```

```
        const password = event.password || ''
        if (!name) return { code: 1, message: '请填写姓名' }
        if (!password || password.length < 4) return { code: 1, message: '密码至少 4
位' }

        const passwordHash = hashPassword(name, password)
        const token = genToken()

        const exist = await users.where({ name }).get()

        if (exist.data.length > 0) {
          const old = exist.data[0]
          if (old.passwordHash) {
            return { code: 1, message: '该姓名已被注册，请直接登录' }
          }
          // 接管小程序旧记录，保留 isAdmin
          await users.doc(old._id).update({ data: { passwordHash, token } })
          return {
            code: 0, token,
            userInfo: { _id: old._id, name, isAdmin: old.isAdmin === true,
avatarFileID: old.avatarFileID || '' }
          }
        }

        const countRes = await users.count()
        const isAdmin = countRes.total === 0

        const addRes = await users.add({
          data: {
            name,
            passwordHash,
            token,
            isAdmin,
            avatarFileID: '',
            createTime: db.serverDate()
          }
        })
        return { code: 0, token, userInfo: { _id: addRes._id, name, isAdmin,
avatarFileID: '' } }
      }

      // ---------- 网页：登录 ----------
      case 'webLogin': {
        const name = (event.name || '').trim()
        const password = event.password || ''
        if (!name || !password) return { code: 1, message: '请填写姓名和密码' }

        const passwordHash = hashPassword(name, password)
        const res = await users.where({ name, passwordHash }).get()
        if (res.data.length === 0) return { code: 1, message: '姓名或密码错误' }

        const user = res.data[0]
        const token = genToken()
        await users.doc(user._id).update({ data: { token } })
        return { code: 0, token, userInfo: cleanUser(user) }
      }

      // ---------- 网页：token 验证 ----------
      case 'getByToken': {
        const { token } = event
        if (!token) return { code: 2, message: '未登录' }
        const res = await users.where({ token }).get()
        if (res.data.length === 0) return { code: 2, message: '登录已过期，请重新登录' }
        return { code: 0, userInfo: cleanUser(res.data[0]) }
```

```
      }

      // ---------- 更新资料（token 或 openid）----------
      case 'updateProfile': {
        let user = null
        if (event.token) {
          const res = await users.where({ token: event.token }).get()
          user = res.data.length > 0 ? res.data[0] : null
        } else if (OPENID) {
          const res = await users.where({ _openid: OPENID }).get()
          user = res.data.length > 0 ? res.data[0] : null
        }
        if (!user) return { code: 2, message: '未登录' }

        const updateData = {}
        if (event.name !== undefined) {
          const n = (event.name || '').trim()
          if (!n) return { code: 1, message: '请填写姓名' }
          updateData.name = n
        }
        if (event.avatarFileID !== undefined) updateData.avatarFileID =
event.avatarFileID
        if (Object.keys(updateData).length > 0) {
          await users.doc(user._id).update({ data: updateData })
        }

        return {
          code: 0,
          userInfo: {
            _id: user._id,
            _openid: user._openid || '',
            name: updateData.name || user.name,
            isAdmin: user.isAdmin === true,
            avatarFileID: updateData.avatarFileID !== undefined ?
updateData.avatarFileID : (user.avatarFileID || '')
          }
        }
      }

      // ---------- 网页：退出 ----------
      case 'logout': {
        const { token } = event
        if (token) {
          const res = await users.where({ token }).get()
          if (res.data.length > 0) await users.doc(res.data[0]._id).update({ data:
{ token: '' } })
        }
        return { code: 0 }
      }

      default:
        return { code: 1, message: '未知操作' }
    }
  } catch (err) {
    console.error(err)
    return { code: 1, message: '服务异常', error: err.message }
  }
}

// ==================== miniprogram/app.js ====================
App({
  onLaunch() {
    if (!wx.cloud) {
      console.error('请使用 2.2.3 或以上的基础库以使用云能力')
    } else {
      wx.cloud.init({
```

```
        env: 'cloud1-d5gjwi2eze187d7e6',
        traceUser: true
      })
    }

    // 启动时检查本地是否已登录过
    const userInfo = wx.getStorageSync('userInfo')
    if (userInfo && userInfo.openid) {
      this.globalData.userInfo = userInfo
    }
  },

  globalData: {
    userInfo: null,
    openid: ''
  }
})

// ==================== miniprogram/pages/admin/admin.js ====================
const app = getApp()
const { formatTime, CATEGORIES } = require('../../utils/util')

Page({
  data: {
    isAdmin: false,
    tab: 'equipment',
    equipmentList: [],
    userList: [],
    recordList: [],
    loading: false,
    showForm: false,
    formMode: 'add', // add / edit
    categories: CATEGORIES,
    categoryIndex: 0,
    form: {
      id: '',
      name: '',
      category: '机身',
      code: '',
      brand: '',
      remark: '',
      imageFileID: ''
    },
    submitting: false
  },

  onLoad() {
    const userInfo = wx.getStorageSync('userInfo')
    const isAdmin = userInfo && userInfo.isAdmin === true
    if (!isAdmin) {
      wx.showToast({ title: '无权限访问', icon: 'none' })
      setTimeout(() => wx.navigateBack(), 1200)
      return
    }
    this.setData({ isAdmin })
  },

  onShow() {
    if (this.data.isAdmin) {
      this.loadEquipment()
      this.loadUsers()
      this.loadRecords()
    }
  },

  switchTab(e) {
```

```
    this.setData({ tab: e.currentTarget.dataset.tab })
  },

  async loadRecords() {
    try {
      const res = await wx.cloud.callFunction({ name: 'borrow', data: { action:
'records' } })
      const result = res.result || {}
      if (result.code === 0) {
        const list = (result.list || []).map(r => ({
          ...r,
          borrowTimeText: r.borrowTime ? formatTime(r.borrowTime) : '',
          returnTimeText: r.returnTime ? formatTime(r.returnTime) : ''
        }))
        this.setData({ recordList: list })
      }
    } catch (e) {
      console.error(e)
    }
  },

  previewReturnImage(e) {
    const url = e.currentTarget.dataset.url
    if (url) wx.previewImage({ urls: [url] })
  },

  previewAvatar(e) {
    const url = e.currentTarget.dataset.url
    if (url) wx.previewImage({ urls: [url] })
  },

  async loadEquipment() {
    try {
      const res = await wx.cloud.callFunction({ name: 'equipment', data: { action:
'list' } })
      const result = res.result || {}
      if (result.code === 0) this.setData({ equipmentList: result.list || [] })
    } catch (e) {
      console.error(e)
    }
  },

  async loadUsers() {
    try {
      const res = await wx.cloud.callFunction({ name: 'admin', data: { action:
'listUsers' } })
      const result = res.result || {}
      if (result.code === 0) this.setData({ userList: result.list || [] })
      else wx.showToast({ title: result.message || '加载失败', icon: 'none' })
    } catch (e) {
      console.error(e)
    }
  },

  openAdd() {
    this.setData({
      showForm: true,
      formMode: 'add',
      categoryIndex: 0,
      form: { id: '', name: '', category: '机身', code: '', brand: '', remark: '',
imageFileID: '' }
    })
  },

  openEdit(e) {
    const item = e.currentTarget.dataset.item
```

```
    const category = item.category || '机身'
    const categoryIndex = Math.max(0, this.data.categories.indexOf(category))
    this.setData({
      showForm: true,
      formMode: 'edit',
      categoryIndex,
      form: {
        id: item._id,
        name: item.name,
        category,
        code: item.code || '',
        brand: item.brand || '',
        remark: item.remark || '',
        imageFileID: item.imageFileID || ''
      }
    })
  },

  onCategoryChange(e) {
    const index = Number(e.detail.value)
    this.setData({ categoryIndex: index, 'form.category':
this.data.categories[index] })
  },

  closeForm() {
    this.setData({ showForm: false })
  },

  noop() {},

  onInput(e) {
    const field = e.currentTarget.dataset.field
    this.setData({ [`form.${field}`]: e.detail.value })
  },

  chooseImage() {
    wx.chooseImage({
      count: 1,
      sizeType: ['compressed'],
      sourceType: ['album', 'camera'],
      success: (res) => {
        const tempPath = res.tempFilePaths[0]
        wx.showLoading({ title: '上传中...' })
        const ext = tempPath.split('.').pop() || 'jpg'
        wx.cloud.uploadFile({
          cloudPath: `equipment/${Date.now()}-${Math.floor(Math.random() *
10000)}.${ext}`,
          filePath: tempPath,
          success: (uploadRes) => {
            this.setData({ 'form.imageFileID': uploadRes.fileID })
            wx.hideLoading()
          },
          fail: () => {
            wx.hideLoading()
            wx.showToast({ title: '上传失败', icon: 'none' })
          }
        })
      }
    })
  },

  removeImage() {
    this.setData({ 'form.imageFileID': '' })
  },

  async submitForm() {
```

```
    const form = this.data.form
    if (!form.name.trim()) {
      wx.showToast({ title: '请填写器材名称', icon: 'none' })
      return
    }

    this.setData({ submitting: true })
    try {
      let res
      if (this.data.formMode === 'add') {
        res = await wx.cloud.callFunction({
          name: 'equipment',
          data: {
            action: 'add',
            name: form.name,
            category: form.category,
            code: form.code,
            brand: form.brand,
            remark: form.remark,
            imageFileID: form.imageFileID
          }
        })
      } else {
        res = await wx.cloud.callFunction({
          name: 'equipment',
          data: {
            action: 'update',
            id: form.id,
            name: form.name,
            category: form.category,
            code: form.code,
            brand: form.brand,
            remark: form.remark,
            imageFileID: form.imageFileID
          }
        })
      }
      const result = res.result || {}
      if (result.code === 0) {
        wx.showToast({ title: this.data.formMode === 'add' ? '添加成功' : '修改成功',
icon: 'success' })
        this.setData({ showForm: false })
        this.loadEquipment()
      } else {
        wx.showToast({ title: result.message || '操作失败', icon: 'none' })
      }
    } catch (e) {
      console.error(e)
      wx.showToast({ title: '网络异常', icon: 'none' })
    } finally {
      this.setData({ submitting: false })
    }
  },

  deleteEquipment(e) {
    const item = e.currentTarget.dataset.item
    wx.showModal({
      title: '删除器材',
      content: `确定删除「${item.name}」吗？此操作不可恢复。`,
      confirmColor: '#e05353',
      success: async (res) => {
        if (!res.confirm) return
        try {
          const call = await wx.cloud.callFunction({
            name: 'equipment',
            data: { action: 'delete', id: item._id }
          })
```

```
          const result = call.result || {}
          if (result.code === 0) {
            wx.showToast({ title: '删除成功', icon: 'success' })
            this.loadEquipment()
          } else {
            wx.showToast({ title: result.message || '删除失败', icon: 'none' })
          }
        } catch (e) {
          wx.showToast({ title: '网络异常', icon: 'none' })
        }
      }
    })
  },

  toggleAdmin(e) {
    const item = e.currentTarget.dataset.item
    const target = !(item.isAdmin === true)
    wx.showModal({
      title: target ? '设为管理员' : '取消管理员',
      content: target ? `确定将「${item.name}」设为管理员吗？` : `确定取消
「${item.name}」的管理员权限吗？`,
      success: async (res) => {
        if (!res.confirm) return
        try {
          const call = await wx.cloud.callFunction({
            name: 'admin',
            data: { action: 'setAdmin', userId: item._id, isAdmin: target }
          })
          const result = call.result || {}
          if (result.code === 0) {
            wx.showToast({ title: result.message, icon: 'success' })
            this.loadUsers()
          } else {
            wx.showToast({ title: result.message || '操作失败', icon: 'none' })
          }
        } catch (e) {
          wx.showToast({ title: '网络异常', icon: 'none' })
        }
      }
    })
  },

  deleteUser(e) {
    const item = e.currentTarget.dataset.item
    wx.showModal({
      title: '移除用户',
      content: `确定移除「${item.name}」吗？移除后该用户将无法再借用器材。`,
      confirmColor: '#e05353',
      success: async (res) => {
        if (!res.confirm) return
        try {
          const call = await wx.cloud.callFunction({
            name: 'admin',
            data: { action: 'deleteUser', userId: item._id }
          })
          const result = call.result || {}
          if (result.code === 0) {
            wx.showToast({ title: result.message, icon: 'success' })
            this.loadUsers()
          } else {
            wx.showToast({ title: result.message || '移除失败', icon: 'none' })
          }
        } catch (e) {
          wx.showToast({ title: '网络异常', icon: 'none' })
        }
      }
```

```
    })
  }
})

// ==================== miniprogram/pages/equipment/equipment.js
====================
const app = getApp()
const { formatTime } = require('../../utils/util')

Page({
  data: {
    id: '',
    item: null,
    userInfo: null,
    isAdmin: false,
    isBorrower: false,
    loading: true,
    operating: false
  },

  onLoad(options) {
    this.setData({ id: options.id || '' })
    const userInfo = wx.getStorageSync('userInfo')
    this.setData({
      userInfo,
      isAdmin: userInfo && userInfo.isAdmin === true
    })
    if (this.data.id) this.loadDetail()
  },

  async loadDetail() {
    this.setData({ loading: true })
    try {
      const res = await wx.cloud.callFunction({
        name: 'equipment',
        data: { action: 'detail', id: this.data.id }
      })
      const result = res.result || {}
      if (result.code === 0 && result.item) {
        const item = result.item
        const userInfo = this.data.userInfo
        this.setData({
          item: {
            ...item,
            borrowTimeText: item.borrowTime ? formatTime(item.borrowTime) : ''
          },
          isBorrower: item.currentBorrowerOpenid && userInfo &&
item.currentBorrowerOpenid === userInfo.openid
        })
      } else {
        wx.showToast({ title: result.message || '加载失败', icon: 'none' })
      }
    } catch (e) {
      console.error(e)
      wx.showToast({ title: '网络异常', icon: 'none' })
    } finally {
      this.setData({ loading: false })
    }
  },

  previewImage() {
    const url = this.data.item && this.data.item.imageFileID
    if (url) wx.previewImage({ urls: [url] })
  },

  requireLogin() {
```

```
  if (this.data.userInfo) return true
  wx.showToast({ title: '请先登录', icon: 'none' })
  setTimeout(() => wx.navigateTo({ url: '/pages/login/login' }), 500)
  return false
},

onBorrow() {
  if (!this.requireLogin()) return
  const item = this.data.item
  wx.showModal({
    title: '借用器材',
    editable: true,
    placeholderText: '请填写借用原因',
    content: `借用「${item.name}」，请填写借用原因：`,
    success: (res) => {
      if (!res.confirm) return
      const reason = (res.content || '').trim()
      if (!reason) {
        wx.showToast({ title: '请填写借用原因', icon: 'none' })
        return
      }
      this.doBorrow(reason)
    }
  })
},

async doBorrow(reason) {
  this.setData({ operating: true })
  try {
    const result = await this.callBorrow('borrow', { reason })
    if (result.code === 0) {
      wx.showToast({ title: '借用成功', icon: 'success' })
      this.loadDetail()
    } else {
      wx.showToast({ title: result.message || '借用失败', icon: 'none' })
    }
  } catch (e) {
    wx.showToast({ title: '网络异常', icon: 'none' })
  } finally {
    this.setData({ operating: false })
  }
},

onReturn() {
  if (!this.requireLogin()) return
  const item = this.data.item
  wx.showModal({
    title: '归还器材',
    content: `归还「${item.name}」前需要拍摄一张照片作为归还凭证，确定继续吗？`,
    success: (res) => {
      if (!res.confirm) return
      this.chooseReturnImage()
    }
  })
},

chooseReturnImage() {
  wx.chooseImage({
    count: 1,
    sizeType: ['compressed'],
    sourceType: ['camera', 'album'],
    success: (res) => {
      const tempPath = res.tempFilePaths[0]
      this.uploadAndReturn(tempPath)
    },
    fail: () => {
```

```
        wx.showToast({ title: '未选择照片，已取消归还', icon: 'none' })
      }
    })
  },

  uploadAndReturn(tempPath) {
    wx.showLoading({ title: '上传中...' })
    const ext = tempPath.split('.').pop() || 'jpg'
    wx.cloud.uploadFile({
      cloudPath: `return/${Date.now()}-${Math.floor(Math.random() * 10000)}.${ext}`,
      filePath: tempPath,
      success: (uploadRes) => {
        wx.hideLoading()
        this.doReturn(uploadRes.fileID)
      },
      fail: () => {
        wx.hideLoading()
        wx.showToast({ title: '上传失败', icon: 'none' })
      }
    })
  },

  async doReturn(returnImageFileID) {
    this.setData({ operating: true })
    try {
      const res = await wx.cloud.callFunction({
        name: 'borrow',
        data: { action: 'return', id: this.data.id, returnImageFileID }
      })
      const result = res.result || {}
      if (result.code === 0) {
        wx.showToast({ title: '归还成功', icon: 'success' })
        this.loadDetail()
      } else {
        wx.showToast({ title: result.message || '归还失败', icon: 'none' })
      }
    } catch (e) {
      wx.showToast({ title: '网络异常', icon: 'none' })
    } finally {
      this.setData({ operating: false })
    }
  },

  callBorrow(action, extra = {}) {
    return wx.cloud.callFunction({
      name: 'borrow',
      data: { action, id: this.data.id, ...extra }
    }).then(res => res.result || {})
  },

  onShareAppMessage() {
    const item = this.data.item
    return {
      title: item ? `器材：${item.name}` : '记者团器材管理',
      path: `/pages/equipment/equipment?id=${this.data.id}`
    }
  }
})

// ==================== miniprogram/pages/index/index.js ====================
const app = getApp()
const { CATEGORIES } = require('../../utils/util')

Page({
  data: {
    userInfo: null,
```

```
    isAdmin: false,
    tab: 'all', // all / available / borrowed
    categories: CATEGORIES,
    selectedCategory: '全部',
    list: [],
    filteredList: [],
    loading: true,
    total: 0,
    availableCount: 0,
    borrowedCount: 0,
    showProfile: false,
    profileName: '',
    profileAvatarFileID: '',
    savingProfile: false
  },

  onShow() {
    const userInfo = wx.getStorageSync('userInfo')
    if (userInfo && userInfo.openid) {
      app.globalData.userInfo = userInfo
      app.globalData.openid = userInfo.openid
      this.setData({
        userInfo,
        isAdmin: userInfo.isAdmin === true
      })
      this.refreshUserInfo()
    } else {
      this.setData({ userInfo: null, isAdmin: false })
    }
    this.loadList()
  },

  // 每次进入首页时从云端刷新最新身份（管理员权限变更后立即生效）
  async refreshUserInfo() {
    try {
      const res = await wx.cloud.callFunction({ name: 'login', data: { action:
'login' } })
      const result = res.result || {}
      if (result.code === 0 && result.isNew) {
        // 用户已被管理员移除，清除本地登录状态
        wx.removeStorageSync('userInfo')
        app.globalData.userInfo = null
        app.globalData.openid = ''
        this.setData({ userInfo: null, isAdmin: false })
        return
      }
      if (result.code === 0 && result.userInfo) {
        const data = {
          openid: result.userInfo._openid,
          _id: result.userInfo._id,
          name: result.userInfo.name,
          avatarFileID: result.userInfo.avatarFileID || '',
          isAdmin: result.userInfo.isAdmin === true
        }
        wx.setStorageSync('userInfo', data)
        app.globalData.userInfo = data
        this.setData({ userInfo: data, isAdmin: data.isAdmin })
      }
    } catch (e) {
      console.error('刷新用户信息失败', e)
    }
  },

  async loadList() {
    this.setData({ loading: true })
    try {
      const res = await wx.cloud.callFunction({
```

```
        name: 'equipment',
        data: { action: 'list' }
      })
      const result = res.result || {}
      if (result.code === 0) {
        const list = result.list || []
        const availableCount = list.filter(i => i.status === 'available').length
        const borrowedCount = list.filter(i => i.status === 'borrowed').length
        this.setData({ list, total: list.length, availableCount, borrowedCount })
        this.applyFilter()
      } else {
        wx.showToast({ title: result.message || '加载失败', icon: 'none' })
      }
    } catch (e) {
      console.error(e)
      wx.showToast({ title: '网络异常', icon: 'none' })
    } finally {
      this.setData({ loading: false })
    }
  },

  switchTab(e) {
    const tab = e.currentTarget.dataset.tab
    this.setData({ tab })
    this.applyFilter()
  },

  switchCategory(e) {
    const selectedCategory = e.currentTarget.dataset.category
    this.setData({ selectedCategory })
    this.applyFilter()
  },

  applyFilter() {
    const { tab, list, selectedCategory } = this.data
    let filteredList = list
    if (tab === 'available') filteredList = filteredList.filter(i => i.status ===
'available')
    if (tab === 'borrowed') filteredList = filteredList.filter(i => i.status ===
'borrowed')
    if (selectedCategory !== '全部') {
      filteredList = filteredList.filter(i => (i.category || '机身') ===
selectedCategory)
    }
    this.setData({ filteredList })
  },

  goDetail(e) {
    const id = e.currentTarget.dataset.id
    wx.navigateTo({ url: '/pages/equipment/equipment?id=' + id })
  },

  goAdmin() {
    wx.navigateTo({ url: '/pages/admin/admin' })
  },

  goRecords() {
    wx.navigateTo({ url: '/pages/records/records' })
  },

  goLogin() {
    wx.navigateTo({ url: '/pages/login/login' })
  },

  openProfile() {
```

```
    const userInfo = this.data.userInfo || {}
    this.setData({
      showProfile: true,
      profileName: userInfo.name || '',
      profileAvatarFileID: userInfo.avatarFileID || ''
    })
  },

  closeProfile() {
    this.setData({ showProfile: false })
  },

  noop() {},

  onProfileNameInput(e) {
    this.setData({ profileName: e.detail.value })
  },

  chooseAvatar() {
    wx.chooseImage({
      count: 1,
      sizeType: ['compressed'],
      sourceType: ['album', 'camera'],
      success: (res) => {
        const tempPath = res.tempFilePaths[0]
        wx.showLoading({ title: '上传中...' })
        const ext = tempPath.split('.').pop() || 'jpg'
        wx.cloud.uploadFile({
          cloudPath: `avatars/${Date.now()}-${Math.floor(Math.random() *
10000)}.${ext}`,
          filePath: tempPath,
          success: (uploadRes) => {
            this.setData({ profileAvatarFileID: uploadRes.fileID })
            wx.hideLoading()
          },
          fail: () => {
            wx.hideLoading()
            wx.showToast({ title: '上传失败', icon: 'none' })
          }
        })
      }
    })
  },

  async saveProfile() {
    const name = this.data.profileName.trim()
    if (!name) {
      wx.showToast({ title: '请填写姓名', icon: 'none' })
      return
    }

    this.setData({ savingProfile: true })
    try {
      const res = await wx.cloud.callFunction({
        name: 'login',
        data: {
          action: 'updateProfile',
          name,
          avatarFileID: this.data.profileAvatarFileID
        }
      })
      const result = res.result || {}
      if (result.code === 0 && result.userInfo) {
        const data = {
          openid: result.userInfo._openid,
          _id: result.userInfo._id,
```

```
          name: result.userInfo.name,
          avatarFileID: result.userInfo.avatarFileID || '',
          isAdmin: result.userInfo.isAdmin === true
        }
        wx.setStorageSync('userInfo', data)
        app.globalData.userInfo = data
        this.setData({ userInfo: data, isAdmin: data.isAdmin, showProfile: false })
        wx.showToast({ title: '保存成功', icon: 'success' })
      } else {
        wx.showToast({ title: result.message || '保存失败', icon: 'none' })
      }
    } catch (e) {
      console.error(e)
      wx.showToast({ title: '网络异常', icon: 'none' })
    } finally {
      this.setData({ savingProfile: false })
    }
  },

  onPullDownRefresh() {
    this.loadList().then(() => wx.stopPullDownRefresh())
  },

  onShareAppMessage() {
    return {
      title: '记者团器材管理',
      path: '/pages/index/index'
    }
  }
})

// ==================== miniprogram/pages/login/login.js ====================
const app = getApp()

Page({
  data: {
    name: '',
    loading: false,
    isNew: true
  },

  onLoad() {
    // 已登录则直接返回上一页
    const userInfo = wx.getStorageSync('userInfo')
    if (userInfo && userInfo.openid) {
      app.globalData.userInfo = userInfo
      app.globalData.openid = userInfo.openid
      this.goBack()
      return
    }

    // 尝试静默登录，判断是否已注册
    this.checkLogin()
  },

  // 返回上一页，若没有上一页则回到首页
  goBack() {
    const pages = getCurrentPages()
    if (pages.length > 1) {
      wx.navigateBack()
    } else {
      wx.reLaunch({ url: '/pages/index/index' })
    }
  },
```

```
  // 静默检查是否已注册
  async checkLogin() {
    try {
      const res = await wx.cloud.callFunction({
        name: 'login',
        data: { action: 'login' }
      })
      const result = res.result || {}
      if (result.code === 0 && !result.isNew && result.userInfo) {
        this.saveAndGo(result.userInfo)
      } else {
        this.setData({ isNew: true })
      }
    } catch (e) {
      console.error('检查登录失败', e)
    }
  },

  onNameInput(e) {
    this.setData({ name: e.detail.value })
  },

  async submit() {
    const name = this.data.name.trim()
    if (!name) {
      wx.showToast({ title: '请填写姓名', icon: 'none' })
      return
    }

    this.setData({ loading: true })
    try {
      const res = await wx.cloud.callFunction({
        name: 'login',
        data: { action: 'register', name }
      })
      const result = res.result || {}
      if (result.code === 0 && result.userInfo) {
        this.saveAndGo(result.userInfo)
      } else {
        wx.showToast({ title: result.message || '登录失败', icon: 'none' })
      }
    } catch (e) {
      console.error(e)
      wx.showToast({ title: '网络异常', icon: 'none' })
    } finally {
      this.setData({ loading: false })
    }
  },

  saveAndGo(userInfo) {
    const data = {
      openid: userInfo._openid,
      _id: userInfo._id,
      name: userInfo.name,
      avatarFileID: userInfo.avatarFileID || '',
      isAdmin: userInfo.isAdmin === true
    }
    wx.setStorageSync('userInfo', data)
    app.globalData.userInfo = data
    app.globalData.openid = data.openid
    this.goBack()
  }
})

// ==================== miniprogram/pages/records/records.js ====================
const app = getApp()
```

```
const { formatTime } = require('../../utils/util')

Page({
  data: {
    recordList: [],
    loading: true,
    isAdmin: false
  },

  onShow() {
    const userInfo = wx.getStorageSync('userInfo')
    if (!userInfo || !userInfo.openid) {
      wx.navigateTo({ url: '/pages/login/login' })
      return
    }
    this.setData({ isAdmin: userInfo.isAdmin === true })
    this.loadRecords()
  },

  async loadRecords() {
    this.setData({ loading: true })
    try {
      const res = await wx.cloud.callFunction({ name: 'borrow', data: { action:
'records' } })
      const result = res.result || {}
      if (result.code === 0) {
        const list = (result.list || []).map(r => ({
          ...r,
          borrowTimeText: r.borrowTime ? formatTime(r.borrowTime) : '',
          returnTimeText: r.returnTime ? formatTime(r.returnTime) : ''
        }))
        this.setData({ recordList: list })
      } else {
        wx.showToast({ title: result.message || '加载失败', icon: 'none' })
      }
    } catch (e) {
      console.error(e)
      wx.showToast({ title: '网络异常', icon: 'none' })
    } finally {
      this.setData({ loading: false })
    }
  },

  previewImage(e) {
    const url = e.currentTarget.dataset.url
    if (url) wx.previewImage({ urls: [url] })
  },

  onPullDownRefresh() {
    this.loadRecords().then(() => wx.stopPullDownRefresh())
  }
})

// ==================== miniprogram/utils/util.js ====================
// 通用工具函数

// 器材分类
const CATEGORIES = ['机身', '镜头', '内存卡', '电池', '闪光灯', '读卡器']

// 调用云函数封装
function callCloud(name, data = {}) {
  return wx.cloud.callFunction({ name, data })
}

// 格式化时间
```

```
function formatTime(date) {
  const d = date instanceof Date ? date : new Date(date)
  const pad = n => (n < 10 ? '0' + n : '' + n)
  return `${d.getFullYear()}-${pad(d.getMonth() + 1)}-${pad(d.getDate())}
${pad(d.getHours())}:${pad(d.getMinutes())}`
}

module.exports = {
  CATEGORIES,
  callCloud,
  formatTime
}

// ==================== miniprogram/pages/admin/admin.wxml ====================
<view class="tabs">
  <view class="tab {{tab === 'equipment' ? 'active' : ''}}" data-tab="equipment"
bindtap="switchTab">器材管理</view>
  <view class="tab {{tab === 'users' ? 'active' : ''}}" data-tab="users"
bindtap="switchTab">用户管理</view>
  <view class="tab {{tab === 'records' ? 'active' : ''}}" data-tab="records"
bindtap="switchTab">借用记录</view>
</view>

<!-- 器材管理 -->
<view wx:if="{{tab === 'equipment'}}" class="container">
  <view class="list-head">
    <text>共 {{equipmentList.length}} 件器材</text>
    <view class="add-btn" bindtap="openAdd">＋ 添加器材</view>
  </view>

  <view wx:if="{{equipmentList.length === 0}}" class="empty">暂无器材，点击右上角添加
</view>

  <view wx:for="{{equipmentList}}" wx:key="_id" class="card admin-card">
    <view class="admin-card-main">
      <view class="admin-card-name">{{item.name}}</view>
      <view class="admin-card-meta">
        <text wx:if="{{item.code}}">编号：{{item.code}}</text>
        <text wx:if="{{item.brand}}">  {{item.brand}}</text>
      </view>
      <view class="status-tag {{item.status === 'available' ? 'available' :
'borrowed'}}">
        {{item.status === 'available' ? '在器材室' : '已借出'}}
      </view>
    </view>
    <view class="admin-card-ops">
      <view class="op op-edit" data-item="{{item}}" bindtap="openEdit">编辑</view>
      <view class="op op-delete" data-item="{{item}}" bindtap="deleteEquipment">删除
</view>
    </view>
  </view>
</view>

<!-- 用户管理 -->
<view wx:if="{{tab === 'users'}}" class="container">
  <view class="list-head">
    <text>共 {{userList.length}} 位用户</text>
  </view>

  <view wx:if="{{userList.length === 0}}" class="empty">暂无用户</view>

  <view wx:for="{{userList}}" wx:key="_id" class="card user-card">
    <view class="user-card-left">
      <image
```

```
        wx:if="{{item.avatarFileID}}"
        class="user-avatar"
        src="{{item.avatarFileID}}"
        mode="aspectFill"
        data-url="{{item.avatarFileID}}"
        bindtap="previewAvatar"
      />
      <view wx:else class="user-avatar user-avatar-text">{{item.name ?
item.name[0] : '?'}}</view>
      <view class="user-card-main">
        <view class="user-name">{{item.name}}</view>
        <view class="user-role">{{item.isAdmin ? '管理员' : '普通成员'}}</view>
      </view>
    </view>
    <view class="user-ops">
      <view class="user-op" data-item="{{item}}" bindtap="toggleAdmin">
        {{item.isAdmin ? '取消管理员' : '设为管理员'}}
      </view>
      <view class="user-op user-op-delete" data-item="{{item}}"
bindtap="deleteUser">移除</view>
    </view>
  </view>
</view>

<!-- 借用记录 -->
<view wx:if="{{tab === 'records'}}" class="container">
  <view class="list-head">
    <text>共 {{recordList.length}} 条记录</text>
  </view>

  <view wx:if="{{recordList.length === 0}}" class="empty">暂无借用记录</view>

  <view wx:for="{{recordList}}" wx:key="_id" class="card record-card">
    <view class="record-top">
      <view class="record-name">{{item.equipmentName}}</view>
      <view class="record-status {{item.status === 'borrowed' ? 'borrowed' :
'returned'}}">
        {{item.status === 'borrowed' ? '借出中' : '已归还'}}
      </view>
    </view>
    <view class="record-line">借用人：{{item.borrowerName}}</view>
    <view class="record-line" wx:if="{{item.reason}}">申请原因：
{{item.reason}}</view>
    <view class="record-line">借用时间：{{item.borrowTimeText}}</view>
    <view class="record-line" wx:if="{{item.status === 'returned'}}">归还时间：
{{item.returnTimeText}}</view>
    <image
      wx:if="{{item.status === 'returned' && item.returnImageFileID}}"
      class="record-img"
      src="{{item.returnImageFileID}}"
      mode="aspectFill"
      data-url="{{item.returnImageFileID}}"
      bindtap="previewReturnImage"
    />
  </view>
</view>

<!-- 添加/编辑器材表单 -->
<view wx:if="{{showForm}}" class="mask" bindtap="closeForm">
  <view class="form-panel" catchtap="noop">
    <view class="form-title">{{formMode === 'add' ? '添加器材' : '编辑器材'}}</view>

    <view class="field">
      <view class="label">器材名称 <text class="req">*</text></view>
```

```
      <input class="input" value="{{form.name}}" data-field="name"
bindinput="onInput" placeholder="如：佳能 5D Mark IV" />
    </view>

    <view class="field">
      <view class="label">分类</view>
      <picker mode="selector" range="{{categories}}" value="{{categoryIndex}}"
bindchange="onCategoryChange">
        <view class="picker">{{form.category}}</view>
      </picker>
    </view>

    <view class="field">
      <view class="label">编号</view>
      <input class="input" value="{{form.code}}" data-field="code"
bindinput="onInput" placeholder="如：CAM-001" />
    </view>

    <view class="field">
      <view class="label">品牌型号</view>
      <input class="input" value="{{form.brand}}" data-field="brand"
bindinput="onInput" placeholder="如：佳能 / EOS 5D4" />
    </view>

    <view class="field">
      <view class="label">图片</view>
      <view class="img-upload">
        <image wx:if="{{form.imageFileID}}" class="preview"
src="{{form.imageFileID}}" mode="aspectFill" bindtap="chooseImage" />
        <view wx:else class="img-add" bindtap="chooseImage">＋</view>
        <view wx:if="{{form.imageFileID}}" class="img-remove"
bindtap="removeImage">移除</view>
      </view>
    </view>

    <view class="field">
      <view class="label">备注</view>
      <textarea class="textarea" value="{{form.remark}}" data-field="remark"
bindinput="onInput" placeholder="备注信息（可选）" auto-height />
    </view>

    <view class="form-btns">
      <button class="btn-cancel" bindtap="closeForm">取消</button>
      <button class="btn-primary" loading="{{submitting}}"
disabled="{{submitting}}" bindtap="submitForm">保存</button>
    </view>
  </view>
</view>

// ==================== miniprogram/pages/equipment/equipment.wxml
====================
<view wx:if="{{loading}}" class="empty">加载中...</view>

<view wx:elif="{{item}}" class="detail">
  <image
    wx:if="{{item.imageFileID}}"
    class="detail-img"
    src="{{item.imageFileID}}"
    mode="aspectFill"
    bindtap="previewImage"
  />
  <view wx:else class="detail-img detail-img-empty">📷</view>
```

```
  <view class="card info-card">
    <view class="name">{{item.name}}</view>
    <view class="status-tag {{item.status === 'available' ? 'available' :
'borrowed'}}">
      {{item.status === 'available' ? '在器材室' : '已借出'}}
    </view>

    <view class="info-row" wx:if="{{item.category}}">
      <view class="info-label">分类</view>
      <view class="info-value">{{item.category}}</view>
    </view>
    <view class="info-row" wx:if="{{item.code}}">
      <view class="info-label">编号</view>
      <view class="info-value">{{item.code}}</view>
    </view>
    <view class="info-row" wx:if="{{item.brand}}">
      <view class="info-label">品牌型号</view>
      <view class="info-value">{{item.brand}}</view>
    </view>
    <view class="info-row" wx:if="{{item.remark}}">
      <view class="info-label">备注</view>
      <view class="info-value">{{item.remark}}</view>
    </view>

    <view wx:if="{{item.status === 'borrowed'}}" class="borrow-info">
      <view class="borrow-title">当前借用信息</view>
      <view class="borrow-line">借用人：{{item.currentBorrower}}</view>
      <view class="borrow-line" wx:if="{{item.currentReason}}">申请原因：
{{item.currentReason}}</view>
      <view class="borrow-line">借用时间：{{item.borrowTimeText}}</view>
    </view>
  </view>

  <view class="actions">
    <button
      wx:if="{{item.status === 'available'}}"
      class="btn-primary action-btn"
      loading="{{operating}}"
      disabled="{{operating}}"
      bindtap="onBorrow"
    >借 用</button>

    <button
      wx:if="{{item.status === 'borrowed' && (isBorrower || isAdmin)}}"
      class="btn-return action-btn"
      loading="{{operating}}"
      disabled="{{operating}}"
      bindtap="onReturn"
    >归 还</button>
  </view>
</view>

// ==================== miniprogram/pages/index/index.wxml ====================
<view class="header">
  <view wx:if="{{userInfo}}" class="user-row" bindtap="openProfile">
    <image wx:if="{{userInfo.avatarFileID}}" class="avatar avatar-img"
src="{{userInfo.avatarFileID}}" mode="aspectFill" />
    <view wx:else class="avatar">{{userInfo.name ? userInfo.name[0] : '?'}}</view>
    <view class="user-info">
      <view class="user-name">{{userInfo.name}}</view>
      <view class="user-role">{{isAdmin ? '管理员' : '普通成员'}} · 点击编辑资料</view>
    </view>
    <view class="header-btn" catchtap="goRecords">📋 记录</view>
    <view wx:if="{{isAdmin}}" class="admin-btn" catchtap="goAdmin">⚙ 管理</view>
  </view>
```

```
  <view wx:else class="user-row" bindtap="goLogin">
    <view class="avatar avatar-guest">访</view>
    <view class="user-info">
      <view class="user-name">未登录</view>
      <view class="user-role">浏览器材无需登录，借用时需登录</view>
    </view>
    <view class="header-btn login-btn">登 录</view>
  </view>
</view>

<view class="tabs">
  <view class="tab {{tab === 'all' ? 'active' : ''}}" data-tab="all"
bindtap="switchTab">全部 ({{total}})</view>
  <view class="tab {{tab === 'available' ? 'active' : ''}}" data-tab="available"
bindtap="switchTab">在器材室 ({{availableCount}})</view>
  <view class="tab {{tab === 'borrowed' ? 'active' : ''}}" data-tab="borrowed"
bindtap="switchTab">已借出 ({{borrowedCount}})</view>
</view>

<scroll-view class="cat-bar" scroll-x enhanced show-scrollbar="{{false}}">
  <view class="cat-item {{selectedCategory === '全部' ? 'active' : ''}}" data-
category="全部" bindtap="switchCategory">全部</view>
  <view
    wx:for="{{categories}}"
    wx:key="*this"
    class="cat-item {{selectedCategory === item ? 'active' : ''}}"
    data-category="{{item}}"
    bindtap="switchCategory"
  >{{item}}</view>
</scroll-view>

<view class="container">
  <view wx:if="{{loading}}" class="empty">加载中...</view>
  <view wx:elif="{{filteredList.length === 0}}" class="empty">
    <view class="empty-icon">📦</view>
    <view>暂无器材</view>
  </view>
  <block wx:else>
    <view
      wx:for="{{filteredList}}"
      wx:key="_id"
      class="card equip-card"
      data-id="{{item._id}}"
      bindtap="goDetail"
    >
      <image
        wx:if="{{item.imageFileID}}"
        class="equip-img"
        src="{{item.imageFileID}}"
        mode="aspectFill"
      />
      <view wx:else class="equip-img equip-img-empty">📷</view>

      <view class="equip-body">
        <view class="equip-name-row">
          <view class="equip-name">{{item.name}}</view>
          <view class="cat-tag">{{item.category || '机身'}}</view>
        </view>
        <view class="equip-meta">{{item.brand}}{{item.brand && item.code ? ' · ' :
''}}{{item.code}}</view>
        <view class="status-row">
          <view class="status-tag {{item.status === 'available' ? 'available' :
'borrowed'}}">
            {{item.status === 'available' ? '在器材室' : '已借出'}}
          </view>
```

```
          <view wx:if="{{item.status === 'borrowed'}}" class="borrower">借用人：
{{item.currentBorrower}}</view>
        </view>
      </view>
    </view>
  </block>
</view>

<!-- 编辑资料弹窗 -->
<view wx:if="{{showProfile}}" class="mask" bindtap="closeProfile">
  <view class="profile-panel" catchtap="noop">
    <view class="form-title">编辑资料</view>

    <view class="avatar-picker" bindtap="chooseAvatar">
      <image wx:if="{{profileAvatarFileID}}" class="avatar-preview"
src="{{profileAvatarFileID}}" mode="aspectFill" />
      <view wx:else class="avatar-preview avatar-placeholder">＋</view>
      <view class="avatar-tip">点击更换头像</view>
    </view>

    <view class="field">
      <view class="label">姓名</view>
      <input class="input" value="{{profileName}}" bindinput="onProfileNameInput"
placeholder="请输入姓名" />
    </view>

    <view class="form-btns">
      <button class="btn-cancel" bindtap="closeProfile">取消</button>
      <button class="btn-primary" loading="{{savingProfile}}"
disabled="{{savingProfile}}" bindtap="saveProfile">保存</button>
    </view>
  </view>
</view>

// ==================== miniprogram/pages/login/login.wxml ====================
<view class="login-wrap">
  <view class="logo-area">
    <view class="logo">📷</view>
    <view class="title">记者团器材管理</view>
    <view class="subtitle">请填写你的姓名进行登录</view>
  </view>

  <view class="form card">
    <view class="form-item">
      <view class="label">姓名</view>
      <input
        class="input"
        type="text"
        placeholder="请输入你的真实姓名"
        placeholder-class="placeholder"
        value="{{name}}"
        bindinput="onNameInput"
      />
    </view>
  </view>

  <button
    class="btn-primary submit-btn"
    loading="{{loading}}"
    disabled="{{loading}}"
    bindtap="submit"
  >登 录</button>

  <view class="tips">登录后将记录你的借用信息，请如实填写姓名</view>
```

```
</view>

// ==================== miniprogram/pages/records/records.wxml ====================
<view class="container">
  <view class="list-head">
    <text>{{isAdmin ? '全部借用记录' : '我的借用记录'}}（{{recordList.length}}）</text>
  </view>

  <view wx:if="{{loading}}" class="empty">加载中...</view>

  <view wx:elif="{{recordList.length === 0}}" class="empty">
    <view class="empty-icon">📋</view>
    <view>暂无借用记录</view>
  </view>

  <block wx:else>
    <view wx:for="{{recordList}}" wx:key="_id" class="card record-card">
      <view class="record-top">
        <view class="record-name">{{item.equipmentName}}</view>
        <view class="record-status {{item.status === 'borrowed' ? 'borrowed' :
'returned'}}">
          {{item.status === 'borrowed' ? '借出中' : '已归还'}}
        </view>
      </view>
      <view class="record-line">借用人：{{item.borrowerName}}</view>
      <view class="record-line" wx:if="{{item.reason}}">申请原因：
{{item.reason}}</view>
      <view class="record-line">借用时间：{{item.borrowTimeText}}</view>
      <view class="record-line" wx:if="{{item.status === 'returned'}}">归还时间：
{{item.returnTimeText}}</view>
      <image
        wx:if="{{item.status === 'returned' && item.returnImageFileID}}"
        class="record-img"
        src="{{item.returnImageFileID}}"
        mode="aspectFill"
        data-url="{{item.returnImageFileID}}"
        bindtap="previewImage"
      />
    </view>
  </block>
</view>

// ==================== miniprogram/app.wxss ====================
page {
  background: #f5f5f5;
  font-size: 28rpx;
  color: #333;
}

.container {
  padding: 24rpx;
}

.btn-primary {
  background: #2b2b2b;
  color: #fff;
  border-radius: 12rpx;
  font-size: 30rpx;
}

.btn-primary[disabled] {
  background: #ccc;
  color: #fff;
}
```

```
.card {
  background: #fff;
  border-radius: 16rpx;
  padding: 24rpx;
  margin-bottom: 24rpx;
  box-shadow: 0 2rpx 12rpx rgba(0, 0, 0, 0.05);
}

// ==================== miniprogram/pages/admin/admin.wxss ====================
.tabs {
  display: flex;
  background: #fff;
  border-bottom: 1rpx solid #eee;
}

.tab {
  flex: 1;
  text-align: center;
  padding: 26rpx 0;
  font-size: 30rpx;
  color: #666;
  position: relative;
}

.tab.active {
  color: #2b2b2b;
  font-weight: bold;
}

.tab.active::after {
  content: '';
  position: absolute;
  bottom: 0;
  left: 50%;
  transform: translateX(-50%);
  width: 48rpx;
  height: 6rpx;
  border-radius: 3rpx;
  background: #2b2b2b;
}

.list-head {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 24rpx 4rpx;
  color: #999;
  font-size: 26rpx;
}

.add-btn {
  background: #2b2b2b;
  color: #fff;
  font-size: 26rpx;
  padding: 12rpx 28rpx;
  border-radius: 30rpx;
}

.empty {
  text-align: center;
  padding: 120rpx 0;
  color: #999;
  font-size: 28rpx;
}
```

```
.admin-card {
  display: flex;
  align-items: center;
}

.admin-card-main {
  flex: 1;
}

.admin-card-name {
  font-size: 30rpx;
  font-weight: bold;
  margin-bottom: 8rpx;
}

.admin-card-meta {
  font-size: 24rpx;
  color: #999;
  margin-bottom: 12rpx;
}

.status-tag {
  display: inline-block;
  font-size: 22rpx;
  padding: 4rpx 16rpx;
  border-radius: 6rpx;
}

.status-tag.available {
  background: #e8f7ee;
  color: #2e9e5b;
}

.status-tag.borrowed {
  background: #fdecec;
  color: #e05353;
}

.admin-card-ops {
  display: flex;
  flex-direction: column;
  align-items: flex-end;
}

.op {
  font-size: 24rpx;
  padding: 8rpx 24rpx;
  border-radius: 24rpx;
  margin: 4rpx 0;
}

.op-edit {
  color: #2b2b2b;
  border: 1rpx solid #999;
}

.op-delete {
  color: #e05353;
  border: 1rpx solid #e05353;
}

.user-card {
  display: flex;
  align-items: center;
  justify-content: space-between;
```

```
}

.user-card-left {
  display: flex;
  align-items: center;
  flex: 1;
  min-width: 0;
}

.user-avatar {
  width: 80rpx;
  height: 80rpx;
  border-radius: 50%;
  background: #f0f0f0;
  margin-right: 20rpx;
  flex-shrink: 0;
  overflow: hidden;
}

.user-avatar-text {
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 34rpx;
  font-weight: bold;
  color: #2b2b2b;
}

.user-name {
  font-size: 30rpx;
  font-weight: bold;
}

.user-role {
  font-size: 24rpx;
  color: #999;
  margin-top: 6rpx;
}

.user-op {
  font-size: 24rpx;
  color: #2b2b2b;
  border: 1rpx solid #999;
  padding: 8rpx 24rpx;
  border-radius: 24rpx;
}

.user-ops {
  display: flex;
  flex-direction: column;
  align-items: flex-end;
  gap: 12rpx;
}

.user-op-delete {
  color: #e05353;
  border-color: #e05353;
}

.record-card {
  padding: 24rpx;
}

.record-top {
  display: flex;
  justify-content: space-between;
```

```
  align-items: center;
  margin-bottom: 12rpx;
}

.record-name {
  font-size: 30rpx;
  font-weight: bold;
}

.record-status {
  font-size: 22rpx;
  padding: 4rpx 16rpx;
  border-radius: 6rpx;
}

.record-status.borrowed {
  background: #fdecec;
  color: #e05353;
}

.record-status.returned {
  background: #e8f7ee;
  color: #2e9e5b;
}

.record-line {
  font-size: 26rpx;
  color: #666;
  line-height: 1.8;
}

.record-img {
  width: 160rpx;
  height: 160rpx;
  border-radius: 12rpx;
  margin-top: 12rpx;
  background: #f0f0f0;
}

/* 表单弹层 */
.mask {
  position: fixed;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  background: rgba(0, 0, 0, 0.5);
  z-index: 100;
  display: flex;
  align-items: flex-end;
}

.form-panel {
  width: 100%;
  background: #fff;
  border-radius: 24rpx 24rpx 0 0;
  padding: 32rpx;
  max-height: 90vh;
  overflow-y: auto;
  box-sizing: border-box;
}

.form-title {
  font-size: 34rpx;
  font-weight: bold;
  text-align: center;
```

```
  margin-bottom: 30rpx;
}

.field {
  margin-bottom: 24rpx;
}

.label {
  font-size: 26rpx;
  color: #666;
  margin-bottom: 12rpx;
}

.req {
  color: #e05353;
}

.input {
  height: 80rpx;
  background: #f5f5f5;
  border-radius: 12rpx;
  padding: 0 20rpx;
  font-size: 28rpx;
}

.picker {
  height: 80rpx;
  line-height: 80rpx;
  background: #f5f5f5;
  border-radius: 12rpx;
  padding: 0 20rpx;
  font-size: 28rpx;
}

.textarea {
  width: 100%;
  background: #f5f5f5;
  border-radius: 12rpx;
  padding: 20rpx;
  font-size: 28rpx;
  box-sizing: border-box;
  min-height: 120rpx;
}

.img-upload {
  display: flex;
  align-items: center;
}

.img-add {
  width: 160rpx;
  height: 160rpx;
  background: #f5f5f5;
  border-radius: 12rpx;
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 60rpx;
  color: #bbb;
}

.preview {
  width: 160rpx;
  height: 160rpx;
  border-radius: 12rpx;
}
```

```
.img-remove {
  margin-left: 24rpx;
  font-size: 26rpx;
  color: #e05353;
}

.form-btns {
  display: flex;
  gap: 24rpx;
  margin-top: 40rpx;
}

.btn-cancel {
  flex: 1;
  background: #f5f5f5;
  color: #666;
  border-radius: 12rpx;
  font-size: 30rpx;
}

.form-btns .btn-primary {
  flex: 1;
  border-radius: 12rpx;
  font-size: 30rpx;
}

// ==================== miniprogram/pages/equipment/equipment.wxss
====================
.empty {
  text-align: center;
  padding: 120rpx 0;
  color: #999;
}

.detail {
  padding: 24rpx;
}

.detail-img {
  width: 100%;
  height: 460rpx;
  border-radius: 16rpx;
  background: #f0f0f0;
}

.detail-img-empty {
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 120rpx;
  color: #ccc;
}

.info-card {
  margin-top: 24rpx;
}

.name {
  font-size: 40rpx;
  font-weight: bold;
  margin-bottom: 20rpx;
}

.status-tag {
```

```
  display: inline-block;
  font-size: 26rpx;
  padding: 8rpx 22rpx;
  border-radius: 8rpx;
  margin-bottom: 20rpx;
}

.status-tag.available {
  background: #e8f7ee;
  color: #2e9e5b;
}

.status-tag.borrowed {
  background: #fdecec;
  color: #e05353;
}

.info-row {
  display: flex;
  padding: 20rpx 0;
  border-bottom: 1rpx solid #f5f5f5;
}

.info-label {
  width: 160rpx;
  color: #999;
  font-size: 28rpx;
}

.info-value {
  flex: 1;
  font-size: 28rpx;
  color: #333;
  word-break: break-all;
}

.borrow-info {
  margin-top: 24rpx;
  padding: 24rpx;
  background: #fdf6f6;
  border-radius: 12rpx;
}

.borrow-title {
  font-size: 28rpx;
  font-weight: bold;
  color: #e05353;
  margin-bottom: 12rpx;
}

.borrow-line {
  font-size: 26rpx;
  color: #666;
  line-height: 1.8;
}

.actions {
  padding: 20rpx 24rpx;
}

.action-btn {
  height: 92rpx;
  line-height: 92rpx;
  font-size: 32rpx;
}
```

```
.btn-return {
  background: #fff;
  color: #e05353;
  border: 1rpx solid #e05353;
  border-radius: 12rpx;
}

// ==================== miniprogram/pages/index/index.wxss ====================
.header {
  background: #2b2b2b;
  padding: 30rpx 32rpx;
}

.user-row {
  display: flex;
  align-items: center;
}

.avatar {
  width: 88rpx;
  height: 88rpx;
  border-radius: 50%;
  background: #fff;
  color: #2b2b2b;
  font-size: 40rpx;
  font-weight: bold;
  display: flex;
  align-items: center;
  justify-content: center;
  margin-right: 24rpx;
}

.avatar-img {
  flex-shrink: 0;
  overflow: hidden;
}

.user-info {
  flex: 1;
}

.user-name {
  color: #fff;
  font-size: 34rpx;
  font-weight: bold;
}

.user-role {
  color: #bbb;
  font-size: 24rpx;
  margin-top: 6rpx;
}

.admin-btn {
  color: #fff;
  font-size: 26rpx;
  padding: 12rpx 24rpx;
  border: 1rpx solid rgba(255, 255, 255, 0.5);
  border-radius: 30rpx;
}

.header-btn {
  color: #fff;
  font-size: 26rpx;
  padding: 12rpx 24rpx;
```

```
  border: 1rpx solid rgba(255, 255, 255, 0.5);
  border-radius: 30rpx;
  margin-right: 16rpx;
}

.login-btn {
  background: #fff;
  color: #2b2b2b;
  border: none;
  font-weight: bold;
}

.avatar-guest {
  background: #555;
  color: #fff;
}

.tabs {
  display: flex;
  background: #fff;
  border-bottom: 1rpx solid #eee;
}

.tab {
  flex: 1;
  text-align: center;
  padding: 26rpx 0;
  font-size: 28rpx;
  color: #666;
  position: relative;
}

.tab.active {
  color: #2b2b2b;
  font-weight: bold;
}

.tab.active::after {
  content: '';
  position: absolute;
  bottom: 0;
  left: 50%;
  transform: translateX(-50%);
  width: 48rpx;
  height: 6rpx;
  border-radius: 3rpx;
  background: #2b2b2b;
}

.cat-bar {
  white-space: nowrap;
  background: #fff;
  padding: 16rpx 24rpx;
  border-bottom: 1rpx solid #eee;
}

.cat-item {
  display: inline-block;
  padding: 10rpx 28rpx;
  margin-right: 16rpx;
  font-size: 26rpx;
  color: #666;
  background: #f5f5f5;
  border-radius: 30rpx;
}
```

```
.cat-item.active {
  background: #2b2b2b;
  color: #fff;
}

.empty {
  text-align: center;
  padding: 120rpx 0;
  color: #999;
  font-size: 28rpx;
}

.empty-icon {
  font-size: 80rpx;
  margin-bottom: 20rpx;
}

.equip-card {
  display: flex;
  padding: 20rpx;
}

.equip-img {
  width: 160rpx;
  height: 160rpx;
  border-radius: 12rpx;
  flex-shrink: 0;
  background: #f0f0f0;
}

.equip-img-empty {
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 60rpx;
  color: #ccc;
}

.equip-body {
  flex: 1;
  margin-left: 24rpx;
  display: flex;
  flex-direction: column;
  justify-content: space-between;
}

.equip-name-row {
  display: flex;
  align-items: center;
}

.equip-name {
  font-size: 32rpx;
  font-weight: bold;
  color: #333;
  margin-right: 16rpx;
}

.cat-tag {
  font-size: 22rpx;
  color: #666;
  background: #f0f0f0;
  padding: 4rpx 14rpx;
  border-radius: 6rpx;
}
```

```
.equip-meta {
  font-size: 26rpx;
  color: #999;
}

.status-row {
  display: flex;
  align-items: center;
}

.status-tag {
  font-size: 24rpx;
  padding: 6rpx 18rpx;
  border-radius: 8rpx;
}

.status-tag.available {
  background: #e8f7ee;
  color: #2e9e5b;
}

.status-tag.borrowed {
  background: #fdecec;
  color: #e05353;
}

.borrower {
  margin-left: 16rpx;
  font-size: 24rpx;
  color: #2e6bd6;
}

/* 编辑资料弹窗 */
.mask {
  position: fixed;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  background: rgba(0, 0, 0, 0.5);
  z-index: 100;
  display: flex;
  align-items: flex-end;
}

.profile-panel {
  width: 100%;
  background: #fff;
  border-radius: 24rpx 24rpx 0 0;
  padding: 32rpx;
  box-sizing: border-box;
}

.form-title {
  font-size: 34rpx;
  font-weight: bold;
  text-align: center;
  margin-bottom: 30rpx;
}

.avatar-picker {
  display: flex;
  align-items: center;
  margin-bottom: 30rpx;
}
```

```
.avatar-preview {
  width: 120rpx;
  height: 120rpx;
  border-radius: 50%;
  background: #f5f5f5;
  overflow: hidden;
}

.avatar-placeholder {
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 60rpx;
  color: #bbb;
}

.avatar-tip {
  margin-left: 24rpx;
  font-size: 26rpx;
  color: #999;
}

.field {
  margin-bottom: 24rpx;
}

.label {
  font-size: 26rpx;
  color: #666;
  margin-bottom: 12rpx;
}

.input {
  height: 80rpx;
  background: #f5f5f5;
  border-radius: 12rpx;
  padding: 0 20rpx;
  font-size: 28rpx;
}

.form-btns {
  display: flex;
  gap: 24rpx;
  margin-top: 40rpx;
}

.btn-cancel {
  flex: 1;
  background: #f5f5f5;
  color: #666;
  border-radius: 12rpx;
  font-size: 30rpx;
}

.form-btns .btn-primary {
  flex: 1;
  border-radius: 12rpx;
  font-size: 30rpx;
}

// ==================== miniprogram/pages/login/login.wxss ====================
.login-wrap {
  padding: 80rpx 40rpx;
}
```

```
.logo-area {
  text-align: center;
  margin-bottom: 80rpx;
}

.logo {
  font-size: 120rpx;
  line-height: 1;
  margin-bottom: 30rpx;
}

.title {
  font-size: 44rpx;
  font-weight: bold;
  color: #2b2b2b;
}

.subtitle {
  margin-top: 16rpx;
  font-size: 28rpx;
  color: #999;
}

.form-item {
  display: flex;
  align-items: center;
}

.label {
  width: 120rpx;
  font-size: 30rpx;
  color: #333;
}

.input {
  flex: 1;
  height: 80rpx;
  font-size: 30rpx;
}

.placeholder {
  color: #bbb;
}

.submit-btn {
  margin-top: 40rpx;
  height: 88rpx;
  line-height: 88rpx;
}

.tips {
  margin-top: 30rpx;
  text-align: center;
  font-size: 24rpx;
  color: #bbb;
}

// ==================== miniprogram/pages/records/records.wxss ====================
.list-head {
  padding: 24rpx 4rpx;
  color: #999;
  font-size: 26rpx;
}
```

```
.empty {
  text-align: center;
  padding: 120rpx 0;
  color: #999;
  font-size: 28rpx;
}

.empty-icon {
  font-size: 80rpx;
  margin-bottom: 20rpx;
}

.record-card {
  padding: 24rpx;
}

.record-top {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 12rpx;
}

.record-name {
  font-size: 30rpx;
  font-weight: bold;
}

.record-status {
  font-size: 22rpx;
  padding: 4rpx 16rpx;
  border-radius: 6rpx;
}

.record-status.borrowed {
  background: #fdecec;
  color: #e05353;
}

.record-status.returned {
  background: #e8f7ee;
  color: #2e9e5b;
}

.record-line {
  font-size: 26rpx;
  color: #666;
  line-height: 1.8;
}

.record-img {
  width: 160rpx;
  height: 160rpx;
  border-radius: 12rpx;
  margin-top: 12rpx;
  background: #f0f0f0;
}

// ==================== miniprogram/app.json ====================
{
  "pages": [
    "pages/index/index",
    "pages/login/login",
    "pages/equipment/equipment",
    "pages/admin/admin",
    "pages/records/records"
```

```
  ],
  "window": {
    "backgroundTextStyle": "dark",
    "navigationBarBackgroundColor": "#2b2b2b",
    "navigationBarTitleText": "记者团器材管理",
    "navigationBarTextStyle": "white",
    "backgroundColor": "#f5f5f5"
  },
  "style": "v2",
  "sitemapLocation": "sitemap.json",
  "lazyCodeLoading": "requiredComponents"
}
```